

SUPRIYO CHAUDHURI

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CORE COMPETENCIES

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|--------------------------------|-------------------------------|---------------------------------------|---------------------------|
| • Data Structures & Algorithms | • Systems | • Distributed Systems | • Technical Communication |
| • Software Engineering | • Artificial Intelligence | • Web Development | • Problem Solving |
| • Database Management | • Object-Oriented Programming | • Global Navigation Satellite Systems | • Scientific Computing |

PROFESSIONAL EXPERIENCE

Space Applications Centre [ISRO]

May 2026 – July 2026

Software Engineering Intern

Ahmedabad, Gujarat

- **Engineered PulsarNav.ai**, a GPS-independent autonomous deep-space navigation platform for spacecraft localization, integrating **X-ray pulsar timing**, celestial reference vectors, recursive state estimation, and scientific simulation to evaluate autonomous navigation across **20+ simulated deep-space mission scenarios** rendering the use of **conventional** Deep Space Navigation (DSN).
- **Designed and implemented** a modular navigation pipeline using the **12.5-year NANOGrav Pulsar Timing Dataset**, processing timing observations from **50+ millisecond pulsars** to develop reusable signal-processing, timing-modeling, spacecraft localization, and performance evaluation frameworks for autonomous navigation research.
- **Implemented** a comprehensive pulsar-based navigation framework inspired by the mathematical formulations presented in **Navigation in Space by X-ray Pulsars** by **Amir Abbas Emadzadeh** and **Jason L. Speyer**, incorporating pulse delay estimation, epoch folding, pulsar navigation geometry, least-squares estimation, Extended Kalman Filtering (EKF), recursive state estimation, spacecraft orbit determination, and pulse timing analysis.
- **Validated** navigation robustness through **1,000+ Monte Carlo simulation trials**, quantifying localization accuracy, estimation uncertainty, and sensitivity to pulsar timing noise across diverse mission scenarios while achieving a **95th-percentile localization error of 6.8 km**.
- **Accelerated** autonomous navigation research and engineering validation by developing an interactive mission analysis platform that visualized spacecraft trajectories, pulsar geometry, navigation telemetry, estimation uncertainty, and algorithm performance while leveraging object-oriented software design, scientific computing, Git-based collaborative development, and reproducible experimentation to deliver a scalable research-grade navigation framework supporting future **DSN**, hybrid navigation, and advanced state-estimation capabilities.

PROJECTS

Expense Tracker | [Node.js](#), [Express.js](#), [PostgreSQL](#), [EJS](#), [JavaScript](#)

December 2025

- **Engineered** a scalable full-stack **personal finance platform** supporting **15+ RESTful APIs**, by designing a modular architecture using **Node.js**, **Express.js**, **PostgreSQL**, and **EJS**.
- **Strengthened** application security by implementing **authentication**, session management, and role-based authorization for secure access to personalized financial records.
- **Optimized** financial data management through a normalized **PostgreSQL** database with **5+ relational tables**, efficient CRUD operations, indexed queries, and relational schema design.
- **Enhanced** financial insights through responsive dashboards supporting **10+ expense categories**, budgeting, transaction history, filtering, and real-time spending analytics using interactive data visualization.
- **Improved** application scalability by optimizing **15+ RESTful APIs**, reducing query latency through database optimization while adopting **Agile**, Git-based version control, debugging, and iterative testing.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, Go, R

Frameworks & Technologies: React.js, Next.js, Fullstack, PostgreSQL, MongoDB, Redis, Git

Developer Tools: VS Code, IntelliJ IDEA Ultimate, Jupyter Notebook, Antigravity, Ollama, Vercel, Obsidian

Certifications: Claude Platform Certification (Anthropic) ; Claude Code & Cowork Certification (Anthropic)

EDUCATION

Pandit Deendayal Energy University

July 2023 – July 2026

B.tech in Computer Engineeering - 3.42 - 4.00

Gandhinagar, Gujarat